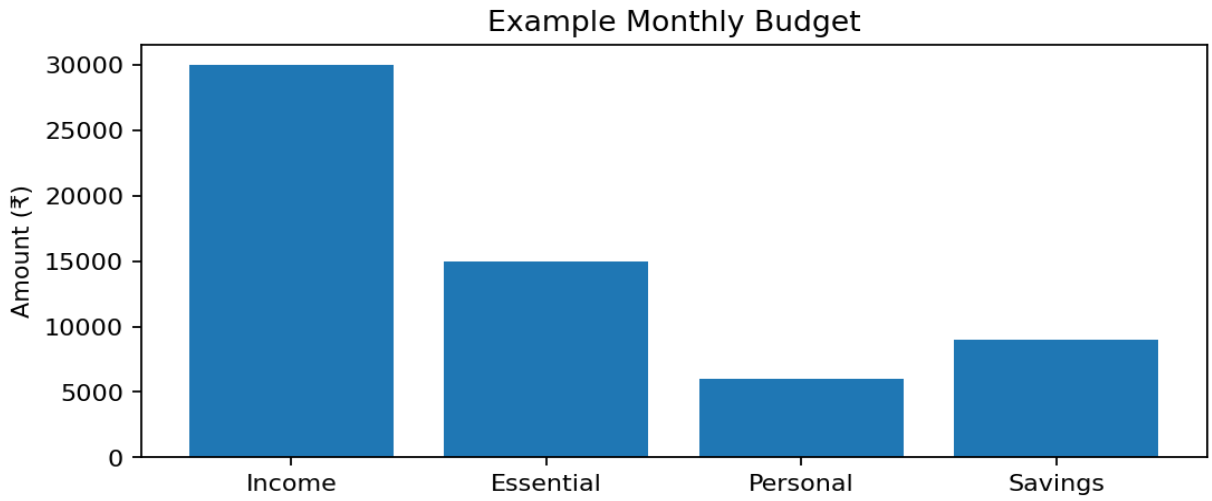
4.1 Smart Budget Planning

Monthly Budget Creation

The Smart Budget Assistant creates a personalized monthly budget by analyzing a user's income, regular expenses, savings target, and financial goals. The purpose is to convert raw financial information into a simple monthly spending plan. Since users have different incomes and responsibilities, the system avoids applying one fixed budget to everyone.

Objectives

- Collect monthly income and recurring expenses.
- Identify essential and non-essential spending.
- Reserve an appropriate amount for savings.
- Show the remaining amount available for flexible spending.
- Provide a clear and manageable monthly plan.



Working Procedure

1. Enter monthly income.
2. Enter fixed expenses such as rent, bills, and transport.
3. Enter a savings goal.
4. Calculate the remaining amount.
5. Display the personalized budget.

Example Data Table

Category	Example	Amount
Monthly Income	Salary / allowance	■30,000
Housing & Bills	Rent, electricity, phone	■8,000

Category	Example	Amount
Food & Essentials	Groceries and daily needs	■7,000
Personal Spending	Shopping / leisure	■5,000
Savings	Future goals	■10,000

Python Implementation

```
income = float(input("Enter monthly income: "))
essential = float(input("Enter essential expenses: "))
personal = float(input("Enter personal spending: "))

savings = income - essential - personal

print("\n--- Monthly Budget ---")
print("Income:", income)
print("Essential Expenses:", essential)
print("Personal Spending:", personal)
print("Savings:", savings)

if savings < 0:
print("Warning: Expenses are higher than income.")
else:
print("Budget created successfully.")
```

Benefits

A personalized budget makes financial planning easier because the plan is based on the user's actual income and expenses. It can also help identify unnecessary spending and maintain a regular savings habit.

Note: All amounts are illustrative examples for academic demonstration. A real application can collect data through forms, a database, or transaction records.